



CONTACT

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Istanbul, Turkey

TECHNICAL SKILLS

PROGRAMMING

C++, C, Python, Java, JavaScript / TypeScript, SQL

SOFTWARE ENG.

OOP, Data Structures, Algorithms, Git, Debugging

WEB & BACKEND

React/Next.js, Node.js, Flask, REST API, PostgreSQL, Docker

AI / ML

Scikit-learn, TensorFlow, OpenCV, RAG, LLM Integration

TEST & QA

Unit/Integration Testing, CI/CD, GitHub Actions, GitLab

SYSTEMS

Linux, TCP/UDP, Real-Time Scheduling, OS

EDUCATION

Istanbul Technical University

Computer Engineering (Senior)
Sep 2021 - Present

Adnan Menderes Anatolian H.S.

2017-2021 | GPA Equiv.: 96.94/100

LANGUAGES

Turkish	Native
English	C1 / Advanced
German	A2

SCHOLARSHIPS & AWARDS

- TUBITAK STAR Research Scholarship
- TEV Academic Excellence Scholarship
- CYDD Merit-Based Scholarship

RELEVANT COURSES

- Data Structures & Algorithms
- Operating Systems
- Software Engineering
- Database Systems
- Computer Organization
- Object-Oriented Programming

INTERESTS

Kickboxing & Fitness | Guitar
Chess | Tech Blogging

BATUHAN KARAKUS

Computer Engineering Student | Junior Software Engineer

C++ · Python · AI/ML · Systems · Backend · Web

PROFILE

Senior Computer Engineering student at Istanbul Technical University (ITU). I build projects across AI, real-time systems, backend and web development using C++, Python, JavaScript/TypeScript and SQL. As a TUBITAK STAR Research Scholar, I gained hands-on experience in ML model training, validation and technical reporting. I prioritize clean code, testable software and problem-driven engineering.

EXPERIENCE

TUBITAK — Research Scholar (Intern)

Feb 2026 – Jun 2026

2247-C STAR Program

- Conducted research on machine learning model training, validation and testing for advanced glycation end-product detection in food samples.
- Managed data preparation, model benchmarking and result reporting pipelines in Python; produced technical research documentation.

TUSAS (Turkish Aerospace) — SKY Experience Remote

Jan 2026 – May 2026

System Avionics & Structural Design

- Completed hands-on work on avionics system architecture, ARINC 653 partitioned system design, and DO-178C software verification approaches.
- Designed test scenarios, applied modular software development and practiced requirement traceability across verification workflows.
- Implemented GitHub Actions-based CI/CD validation pipelines; produced technical reports using LaTeX.

Teknosa — Step 2025 Program

Jul 2025

Web/UX & Frontend Developer

- Managed banner, category and product content workflows on SAP Commerce Cloud (Hybris) and SmartEdit CMS in a corporate e-commerce environment.
- Analyzed product data gaps using SQL and Python/Pandas; produced reporting dashboards with Power BI.
- Contributed to user flow analysis, bug reporting and HTML/CSS interface adjustments across e-commerce pages.

PROJECTS

FPMC-Avionics-RealTime

C++ · GoogleTest · GitHub Actions

Deterministic C++ avionics flight-phase monitoring component with 77 test cases covering boundary-value, fault-handling and regression scenarios; zero-allocation circular buffer and microbenchmark-based performance validation.

local-wikipedia-rag-assistant

Python · ChromaDB · Ollama · RAG

Locally running RAG assistant over Wikipedia data; covers embedding generation, ChromaDB-based retrieval pipeline and Ollama LLM integration.

gcs-autopilot-commsys

C++11 · TCP/UDP · Integration Testing

GCS-autopilot command/telemetry communication prototype with checksum validation, heartbeat watchdog, XOR integrity checks and system integration tests.

SAM-Assistive-Vision

Python · Computer Vision · AI Pipeline

AI-powered visual assistant for visually impaired users; focuses on segmentation, classification and description generation pipelines.

Distributed Microservice-Based E-Commerce

Node.js · Express · Go/gRPC · Docker

Microservice-based backend prototype focusing on inter-service communication, API design and containerized development workflows.

Data Structures Repository

C++ · Algorithms · GitHub Actions

Linked list, tree, heap and graph implementations with GitHub Actions-based build and test automation pipeline.

CERTIFICATIONS & TRAINING

- 2026 DO-178C Avionics Software Training — TUSAS SKY Experience
- 2025 Cybersecurity Fundamentals Course (In Progress)
- 2025 Object-Oriented Java Programming: Data Structures — Coursera
- 2024 Project Management & Scrum Fundamentals Certificate
- 2023 Web Development Fundamentals (HTML, CSS, JS) — Google Developer Courses
- 2022 Data Science & Machine Learning with Python Certificate
- 2022 Robotics Coding Certificate — ITU